# ELL7286 / ELL784 Assignment 1
# Written Component

## Problem 1: Machine Learning Paradigms

(a) • Stage **1 — Unsupervised Learning:**
The sensor measurements are **unlabeled**, so there is no target variable telling the model what each measurement represents. The system analyzes the data to discover recurring patterns or groups of similar environmental conditions. A suitable technique would be **clustering**, such as K-means.

• Stage **2 — Supervised Learning:**
Here, experts provide labels such as **normal** or **hazardous** for a subset of measurements. The model learns a mapping from sensor measurements (features) to the corresponding labels and then predicts the condition of new measurements.
Since there are two possible classes, this is a **binary classification** problem.
For example, logistic regression, decision trees, SVM, or neural networks could be used.

• Stage **3 — Reinforcement Learning:**
The mobile sensor learns **which actions to take** through interaction with the environment. It receives rewards for early hazard detection and penalties for movement and energy consumption, allowing it to learn a policy that maximizes cumulative reward.

(b) Only **Stage 2** uses supervised learning.

| **Component** | **Stage 2** |
|---|---|
| **Input** | Sensor measurements/features from a new environmental observation |
| **Target** | The expert-provided label: **normal** or **hazardous** |
| **Model output** | Predicted environmental condition: **normal** or **hazardous** |
| **Task type** | **Binary classification** |

The model learns a mapping from the sensor measurements to the corresponding expert-provided labels and uses this mapping to classify new measurements.

(c) In **Stage 1**, no labeled targets are provided. The system learns **patterns and structure present in the sensor measurements**, such as groups of similar measurements representing recurring environmental conditions. For example, it may learn clusters corresponding to different types of environmental states without being explicitly told what those states mean.
It learns the **underlying structure/patterns in the data**, rather than learning to predict a predefined target.

(d) **System A is not a machine-learning system**, because its prediction rule and the threshold are both **fixed by an expert**. It does not learn the rule or its parameters from data.

**System B is a machine-learning system**, because although it uses the same simple threshold rule for prediction, the **threshold is learned automatically from historical labeled data** by minimizing classification errors.

The key distinction is that **the prediction rule itself need not be complex for a system to use machine learning**. What makes System B a machine-learning system is the **data-driven process used to determine the rule's parameter (the threshold)**.

## Problem 2: Interpreting a Linear Regression Model

**(a)** The model

$$\hat{y} = \theta_0 + \theta_1 T + \theta_2 P + \theta_3 TP$$

is still a **linear regression model because it is linear in the unknown parameters** $\theta_0$, $\theta_1$, $\theta_2$, and $\theta_3$. Although TP is a product of the input variables, it can be treated as an additional feature in the regression model.

The model can therefore be written as:

$$\hat{y} = \theta_0(1) + \theta_1(T) + \theta_2(P) + \theta_3(TP)$$

which is a linear combination of the features 1, T, P, and TP. Thus, linearity in the parameters, rather than necessarily in the original input variables, determines whether it is a linear regression model.

**(b)** The interpretation of $\theta_1$ as the change in the predicted response for a one-unit increase in temperature is **incomplete** because the model contains the interaction term TP. Therefore, the effect of temperature depends on the value of pressure P.

When T increases by one unit while P is held fixed:

Original prediction:

$$\hat{y} = \theta_0 + \theta_1 T + \theta_2 P + \theta_3 TP$$

New prediction:

$$\hat{y}_new = \theta_0 + \theta_1(T + 1) + \theta_2 P + \theta_3(T + 1)P$$

The change in the predicted response is:

$$\Delta\hat{y} = \hat{y}_new - \hat{y}$$

$$\Delta\hat{y} = \theta_1 + \theta_3 P$$

Therefore, the effect of increasing temperature by one unit is $\boldsymbol{\theta_1 + \theta_3 P}$, not simply $\theta_1$. This shows that the effect of temperature depends on the fixed pressure P.

(c) Since **T_F = 1.8T + 32**, the features **1, T, and T_F are linearly dependent**. In particular,

$$\mathbf{T_F = 1.8T + 32(1)}.$$

Therefore, the design matrix does not have full column rank, so the least-squares parameter vector may **not be unique**.

Yes, **different parameter vectors can still produce exactly identical predictions**. This is because different combinations of the coefficients of the intercept, T, and T_F can represent the same relationship with the response.

Feature cleaning or preprocessing should be performed before fitting the model. Specifically, **one of the redundant features, T or T_F, should be removed**. This eliminates the perfect multicollinearity and allows the model to have a unique parameter vector while preserving the information contained in temperature.

## Problem 3: Weighted Linear Regression

(a) Let the training set contain N samples, with weight $w_n$ assigned to the n-th sample. The empirical risk is

$$R(\theta) = 1/2 \, \Sigma_{n=1}^{N} w_n(\theta^T x^{(n)} - y^{(n)})^2.$$

Define the residual vector as:

$$r = X\theta - y.$$

The weighted squared residual can be written in matrix form as:

$$R(\theta) = 1/2 \, r^T W r$$

Therefore,

$$\mathbf{R(\theta) = 1/2 \, (X\theta - y)^T W (X\theta - y).}$$

Here, **W is a diagonal weight matrix** containing the weight of each training sample:

$$\mathbf{W = diag(w_1, w_2, ..., w_N).}$$

Thus, W has dimensions **N × N**, while X is the design matrix containing the N training samples and y is the N-dimensional target vector.

The diagonal structure of W ensures that each squared residual is multiplied by its corresponding weight $w_n$.

(b) For the n-th training sample,

$$y^{(n)} = \theta^T x^{(n)} + \varepsilon^{(n)},$$

and

$$\varepsilon^{(n)} \sim N(0, \sigma_n^2).$$

Therefore, conditioned on $x^{(n)}$, the target follows a Gaussian distribution:

$$y^{(n)} \mid x^{(n)}, \theta \sim N(\theta^T x^{(n)}, \sigma_n^2).$$

The likelihood of the complete training dataset, using independence of the noise variables, is

$$L(\theta) = \prod_{n=1}^{N} [1/(\sqrt{(2\pi)}\sigma_n)] \exp[-(y^{(n)} - \theta^T x^{(n)})^2/(2\sigma_n^2)].$$

Taking the logarithm,

$$\log L(\theta) = -\Sigma_{n=1}^{N} [\log(\sqrt{(2\pi)}\sigma_n) + (y^{(n)} - \theta^T x^{(n)})^2/(2\sigma_n^2)].$$

Since $\sigma_n^2$ is known, the first term does not depend on θ. Therefore, maximizing the likelihood is equivalent to minimizing

$$\Sigma_{n=1}^{N} (1/(2\sigma_n^2))(\theta^T x^{(n)} - y^{(n)})^2.$$

This has the same form as the weighted least-squares objective

$$R(\theta) = 1/2 \Sigma_{n=1}^{N} w_n(\theta^T x^{(n)} - y^{(n)})^2.$$

Hence, the required weight is

$$\mathbf{w_n = 1/\sigma_n^2.}$$

Therefore, samples with **smaller noise variance receive larger weights**, because their observations are more reliable.

**Final result:**

$$\mathbf{w_n = 1/\sigma_n^2}.$$

## Problem 4: Logistic Regression with Perfectly Separable Data

(a) For logistic regression,

$$\mathbf{f(x^{(n)}; \theta) = \sigma(\theta^T x^{(n)}) = 1 / (1 + e^{-z}),}$$

$$\mathbf{where\ z = \theta^T x^{(n)}.}$$

The sigmoid function satisfies:

$$\mathbf{\sigma(z) > 1/2\ when\ z > 0,}$$

$$\mathbf{\sigma(z) < 1/2\ when\ z < 0.}$$

Since the data are perfectly separable, we have**:**

- **When $y^{(n)} = 1$, $\theta^T x^{(n)} > 0$. Therefore, $f(x^{(n)}; \theta) > 1/2$.**
- **When $y^{(n)} = 0$, $\theta^T x^{(n)} < 0$. Therefore, $f(x^{(n)}; \theta) < 1/2$.**

Hence, the separability conditions guarantee that the logistic model correctly classifies every training sample using the threshold 1/2.

(b) From part (a), because the training data are perfectly separable:

- **For every sample with $y^{(n)} = 1$, we have $f(x^{(n)}; \theta) > 1/2$.**
- **For every sample with $y^{(n)} = 0$, we have $f(x^{(n)}; \theta) < 1/2$, so $1 - f(x^{(n)}; \theta) > 1/2$.**

The likelihood is

$$\mathbf{L(\theta) = \prod_{n=1}^{N} f(x^{(n)}; \theta)^{y(n)} [1 - f(x^{(n)}; \theta)]^{1-y(n)}.}$$

There are $N_1$ samples belonging to class 1. Each of their likelihood contributions is greater than 1/2. Similarly, there are $N_0$ samples belonging to class 0, and each of their contributions $[1 - f(x^{(n)}; \theta)]$ is greater than 1/2.

Therefore,

$$\mathbf{L(\theta) > (1/2)^{N1}\ (1/2)^{N0}.}$$

Since $N_1 + N_0 = N$**,**

$$L(\theta) > (1/2)^{N_1+N_0}$$

$$L(\theta) > (1/2)^{N}.$$

Thus,

$$L(\theta) > (1/2)^{N}.$$

(c) Let $z^{(n)} = \theta^T x^{(n)}$.

Because the data are perfectly separable:

- **If $y^{(n)} = 1$, then $z^{(n)} > 0$.**
- **If $y^{(n)} = 0$, then $z^{(n)} < 0$.**

**For $\alpha > 1$:**

**Case 1: $y^{(n)} = 1$**

**Since $z^{(n)} > 0$,**

**$\alpha z^{(n)} > z^{(n)}$.**

The sigmoid function $\sigma(z)$ is strictly increasing, so

**$\sigma(\alpha z^{(n)}) > \sigma(z^{(n)})$.**

Therefore, the likelihood contribution from every positive sample increases.

**Case 2: $y^{(n)} = 0$**

**Since $z^{(n)} < 0$,**

**$\alpha z^{(n)} < z^{(n)}$.**

Because $\sigma(z)$ is increasing,

**$\sigma(\alpha z^{(n)}) < \sigma(z^{(n)})$.**

Hence,

**$1 - \sigma(\alpha z^{(n)}) > 1 - \sigma(z^{(n)})$.**

Therefore, the likelihood contribution from every negative sample also increases.

Since every individual likelihood contribution increases**,**

**$L(\alpha\theta) > L(\theta)$, for every $\alpha > 1$.**

Thus, scaling the separating parameter vector by a factor greater than 1 always increases the likelihood.

(d) For every training sample, the data are perfectly separable.

- **If $y^{(n)} = 1$, then $\theta^T x^{(n)} > 0$. As $\alpha \to \infty$,**
  **$\sigma(\alpha\theta^T x^{(n)}) \to 1$.**
- **If $y^{(n)} = 0$, then $\theta^T x^{(n)} < 0$. As $\alpha \to \infty$,**
  **$\sigma(\alpha\theta^T x^{(n)}) \to 0$,**
  **and therefore**
  **$1 - \sigma(\alpha\theta^T x^{(n)}) \to 1$.**

Hence, every individual likelihood contribution approaches 1. Therefore,

$\lim_{\alpha\to\infty} L(\alpha\theta) = 1$.

However, this value is not attained by any finite parameter vector, because for any finite value of $\alpha$, the sigmoid function produces probabilities strictly between 0 and 1.

Therefore, the likelihood can approach 1 but never actually reaches 1. Consequently, the maximum-likelihood estimate does not exist as a finite parameter vector. Hence, there is no finite unique MLE; the parameter norm can grow without bound while the likelihood continues to increase toward its supremum of 1.

**Conclusion: Perfect separation causes logistic regression's maximum-likelihood parameters to diverge rather than converge to a finite solution.**

## Problem 5: Classification Thresholds and Evaluation

(a) formulas are:

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1-score = 2 × Precision The × Recall / (Precision + Recall)

For threshold $c_1$:
TP = 18, FP = 80, TN = 900, FN = 2.

Accuracy:

Accuracy = (18 + 900) / 1000
= 918 / 1000
= 0.918

Precision:

Precision = 18 / (18 + 80)
= 18 / 98
≈ 0.1837

Recall:

Recall = 18 / (18 + 2)
= 18 / 20
= 0.9000

F1-score:

F1 = 2 × (0.1837 × 0.9000) / (0.1837 + 0.9000)
≈ 0.3051

Therefore, for $c_1$:

- Accuracy = 0.9180
- Precision = 0.1837
- Recall = 0.9000
- F1-score = 0.3051

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For threshold $c_2$:
TP = 12, FP = 20, TN = 960, FN = 8.

Accuracy:

Accuracy = (12 + 960) / 1000
= 972 / 1000
= 0.972

Precision:

Precision = 12 / (12 + 20)
= 12 / 32
= 0.3750

Recall:

Recall = 12 / (12 + 8)
= 12 / 20
= 0.6000

F1-score:

F1 = 2 × (0.3750 × 0.6000) / (0.3750 + 0.6000)
≈ 0.4615

Therefore, for $c_2$:

- Accuracy = 0.9720
- Precision = 0.3750
- Recall = 0.6000
- F1-score = 0.4615

Final comparison

| Metric | $c_1$ | $c_2$ |
|---|---|---|
| Accuracy | 0.9180 | 0.9720 |
| Precision | 0.1837 | 0.3750 |
| Recall | 0.9000 | 0.6000 |
| F1-score | 0.3051 | 0.4615 |

Notice that $c_1$ has much higher recall, while $c_2$ has better precision, F1-score, and accuracy.

(b) I would prefer threshold $c_1$ because failing to detect a machine failure is more costly than issuing a false alarm. Threshold $c_1$ has a recall of 0.90, with only 2 false negatives, compared with a recall of 0.60 and 8 false negatives for $c_2$.

Although $c_1$ produces more false positives (80 compared with 20), the higher recall means that it detects more actual machine failures. Since false negatives are substantially more costly, maximizing recall is more important in this application.
Therefore, threshold $c_1$ is preferred.

(c) Accuracy can be misleading because machine failure is a rare event, so most of the test samples are non-failures. As a result, a model can achieve high accuracy mainly by correctly predicting the large number of normal cases while still missing a significant number of actual failures.

In this application, false negatives are much more costly than false positives. Therefore, metrics such as recall, precision, and F1-score are more informative than accuracy alone when evaluating the ability to detect machine failures.